

TPS/CLM Platform Planning & Installation

LK02-200

**Communication Link Module
(CLM)**

***TPS/CLM Platform
Planning & Installation***

**LK02-200
Release Independent
03/2013**

Copyright, Notices, and Trademarks

Printed in U.S.A. – © Copyright 1999-2013 by Honeywell International Srl.

Revision 1.3 – March 2013

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customer.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

TotalPlant Solution (**TPS**) System is a U. S. registered trademark of Honeywell Inc. TPS is the evolution of TDC 3000.

TPS is the evolution of TDC 3000

Local Control Network, and LCN are trademarks of Honeywell Inc.

TPS Network (**TPN**) is the LCN/UCN portion of the **TotalPlant** Solution System.

Microsoft, Windows, Windows 2000, and Windows 2003/2008 are trademarks of Microsoft Corporation.

This document was prepared using Microsoft Word for Windows.

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA
1-800 822-7673

About This Publication

This publication provides the general description, site preparation, installation, and checkout instructions for the APP/CLM.

For an installation to a Communication Link Module (CLM), which is hosted on an AM with CLM communications boards (CLMI, CCP, UCIM) installed, refer to the *CLM Planning, Installation and Service Manual LK02-100*.

Support and Other Contacts

United States and Canada

Contact: Honeywell Solution Support Center
Phone: 1-800-822-7673
Calls are answered by dispatcher between 6:00 am and 4:00 pm Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.
Fascimile: 1-973-455-5000
Mail: Honeywell TAC, MS L17
1860 W. Rose Garden Lane
Phoenix, AZ, 85027 USA

Europe, Middle East, and Africa (EMEA)

Contact: Honeywell TAC – EMEA
Phone: +32-2-728-2345
Fascimile: +32-2-728-2696
Mail: TAC-BE02
Hermes Plaza
Hermeslaan, 1H
B-1831 Diegem, Belgium
Email: tac-be02@honeywell.com

Pacific

Contact: Honeywell Global TAC – Pacific
Phone: 1300-364-822 (toll free within Australia)
+61-8-9362-9559 (outside Australia)
Fascimile: +61-8-9362-9564
Mail: Honeywell Limited Australia
5 Kitchener Way
Burswood, WA, 6100, Australia
Email: GTAC@honeywell.com

India

Contact: Honeywell Global TAC – India
Phone: +91-20- 6603-2718/19
1800-233-5051
Fascimile: +91-20- 6603-9800
Mail: Honeywell Automation India Ltd
56 and 57, Hadapsar Industrial Estate
Hadapsar, Pune –411 013, India
Email: Global-TAC-India@honeywell.com

Korea

Contact: Honeywell Global TAC – Korea
Phone: +82-80-782-2255 (toll free within Korea)
Fascimile: +82-2-792-9015
Mail: Honeywell Co., Ltd
4F, Sangam IT Tower B4-4 Block
1590, DMC Sangam-dong, Mapo-gu
Seoul, 121-835, Korea
Email: Global-TAC-Korea@honeywell.com

People's Republic of China

Contact: Honeywell Global TAC – China
Phone: +86- 21-2219-6888
800-820-0237
400-820-0386
Mail: Honeywell (China) Co., Ltd
33/F, Tower A, City Center, 100 Zunyi Rd.
Shanghai 200051, People's Republic of China
Email: Global-TAC-China@honeywell.com

Singapore

Contact: Honeywell Global TAC – South East Asia
Phone: +65-6823-2215
Fascimile: +65-6445-3033
Mail: Honeywell Private Limited
Honeywell Building
17, Changi Business Park Central 1
Singapore 486073
Email: GTAC-SEA@honeywell.com

Japan

Contact: Honeywell Global TAC – Japan
Fascimile: +81-3-6730-7228
Mail: Honeywell K.K
New Pier Takeshiba, South Tower Building
20th Floor, 1-16-1 Kaigan, Minato-ku
Tokyo 105-0022, Japan
Email: Global-TAC-JapanJA25@honeywell.com

Elsewhere

Call your nearest Honeywell office.

World Wide Web

Honeywell Process Solutions website:

<https://www.honeywellprocess.com/>

Training Classes

Honeywell Automation College:

<http://www.automationcollege.com>

Table of Contents

1. INTRODUCTION	13
1.1. General Description	13
1.2. CLM Architecture	14
1.2.1. AM-based CLM - Hosted on an Application Module	14
1.2.2. APP/CLM	15
1.3. Honeywell Support Services.....	16
1.4. Related Publications.....	16
2. SITE PREPARATION	17
2.1. Site Requirements	17
2.2. Configuration	17
2.2.1. Cabling Considerations	17
2.2.1.1. Cable Lengths.....	17
2.2.1.2. Direct Connection.....	17
2.2.1.3. Typical cabling connections	18
2.2.1.4. Cable connections for optional port expanders	20
3. CLM PLATFORM SOFTWARE INSTALLATION.....	21
3.1. Introduction.....	21
3.1.1. APP-based Installation	21
3.2. APP/CLM Software – The Different Pieces	21
3.2.1. APP/CLM Installation Overview	22
3.2.1.1. Installation overview for TPS APP.....	22
3.2.1.2. Installation overview for Experion APP	22
3.3. Windows 2003 Only - Install the TPS-Based Software to the APP/CLM.....	24
3.3.1. Windows 2003 Only - Shutdown TDC Emulation Service	24
3.3.1.1. Shutdown TDC Emulators Service.....	24
3.3.2. Windows 2003 Only - Install TPS/CLM Windows 2003-Based Software	26
3.3.2.1. Install the Windows 2003-based Software	26
3.3.3. Windows 2003 Only - Configure the Emulated Board(s).....	26
3.3.3.1. Configure the Logical Boards.....	26
3.3.4. Windows 2003 Only - Start TDC Emulation Service	28
3.3.4.1. Start TDC Emulators Service.....	28
3.3.4.2. The next step	30
3.4. Windows 2008 Only - Install the Experion-Based Software to Experion APP.....	31
3.4.1. Windows 2008 Only - Shutdown TDC Emulation Service	31
3.4.1.1. Shutdown TDC Emulators Service.....	31
3.4.2. Windows 2008 Only - Install APP/CLM Windows 2008-Based Software.....	33
3.4.2.1. Install the Windows 2008-based Software	33
3.4.3. Windows 2008 Only - Configure the Emulated Board(s).....	34
3.4.3.1. Configure the Logical Boards.....	34
3.4.4. Windows 2008 Only - Start TDC Emulation Service	36

3.4.4.1.	Start TDC Emulators Service.....	36
3.4.4.2.	The next step	38
3.5.	Install the LCN-Based Software to the APP/CLM	39
3.5.1.	Overview.....	39
3.5.1.1.	Configuration overview.....	39
3.5.2.	Install CLM Platform Software.....	40
3.5.2.1.	Before You Do the Software Installation	40
3.5.3.	Execute the Install Procedure	41
3.5.4.	Build Required CLM Platform Points.....	42
3.5.4.1.	Build the Required CLM_nn & BUILD_nn LCN Points	42
3.5.5.	Customize Displays.....	44
3.5.5.1.	Custom Graphics Must be Compiled.....	44
3.5.6.	Configure the Emulated CLMI Board.....	46
3.5.6.1.	You Must Define the Emulated Board Location	46
3.5.7.	Configure the Emulated CLMI Board, continued.....	47
3.5.7.1.	Edit the Configuration File	47
3.5.8.	Update LCN Network Configuration File	48
3.5.8.1.	NCF update process	48
3.5.8.2.	Always Backup NCF Before Making Changes	48
3.5.8.3.	Modify NCF	49
3.5.9.	Restart the APP with CLM Platform.....	52
3.5.10.	Adding Other APP/CLM after the First Installation.....	53
3.5.10.1.	Adding other CLMs.....	53
3.6.	Install CLM Application.....	54
4.	INTERFACE ERRORS	55
5.	APPENDIX – UCLM: CCP AND UCIM REDESIGNED	57
5.1.	Ordering Information.....	57
5.2.	Switch Settings of UCLM board.....	57

Acronyms

AM.....	TPN Application Module
APP.....	Application Processing Platform
CCP	Communication & Control Processor Board
CDS	Custom Data Segment
CL.....	Control Language
CLM	Communication Link Module
CLMI	Communication Link Module Interface
EB.....	Exception Build
GUS	Global User Station
HM.....	TPS History Module
LCN	TPS Local Control Network
NCF	Network Configuration File
NDT	Network Definition Table
TDF	Template Definition File
TPN	TotalPlant Network
TPS	TotalPlant Solution System
US.....	Universal Station
W2K	Microsoft Windows 2000 Operating System
EAPP.....	Experion APP

1. INTRODUCTION

1.1. General Description

The APP/CLM runs in a Honeywell TPS and Experion node with Application Module; (AM) personality software installed. It is important to note that the APP/CLM requires the TPS and the Experion node to be configured as an “AM Replacement” node, and is expected to be normally configured as an APP. This distinction is made in the Configuration Utility during the definition of the LCNP4 board when selecting AM vs. APP as the type.

Note! For clarity, this manual will refer to the TPS node and Experion node as the APP, but keep in mind that it may actually be configured as an AM replacement node only.

The APP/CLM provides a method of connecting various non-Honeywell physical devices (such as Modbus RS-232 and 485 based devices) and logical devices (such as OPC servers) to the Honeywell system.

The two primary software segments included in this package are: (1) the CLM software loaded to the Application Module (Native Window) side to create the CLM Platform and, (2) the Windows-based software loaded to the TPS and Experion node to support CLM communications and data transfer to other physical or logical devices.

Important! With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32 bit and 64 bit variants). For more information, see the section: *Windows 2008 Only - Install the Experion-Based Software to the Experion APP*

No specific special hardware is provided with the APP/CLM.

A Note about Physical and Emulated CLM Interface boards:

Earlier CLM Platform packages that were installed in the AM node required CLMI, CCP, or UCIM boards. The concept of these boards and communications ports is carried over and emulated in the APP/CLM. However, the communications and application capability is now supported solely in the Windows environment, and include either physical or logical communications devices. The actual device is transparent to the CLM operation.

This manual guides you through the planning and installation considerations unique to the APP/CLM. However, this manual is not intended to be a substitute for either the *LCN Site Planning* manual or the *LCN System Installation* manual.

1.2. CLM Architecture

Sections 1.2.1 and 1.2.2 following depict the architectural overview of the two current offerings of the CLM, the AM-based CLM and the APP/ CLM. Review the figures in each section for comparison of these different, yet similar, offerings.

1.2.1. AM-based CLM - Hosted on an Application Module

The following figure depicts the architecture of the Communications Link Module hosted on an Application Module. *This figure is presented for comparison reference only!*

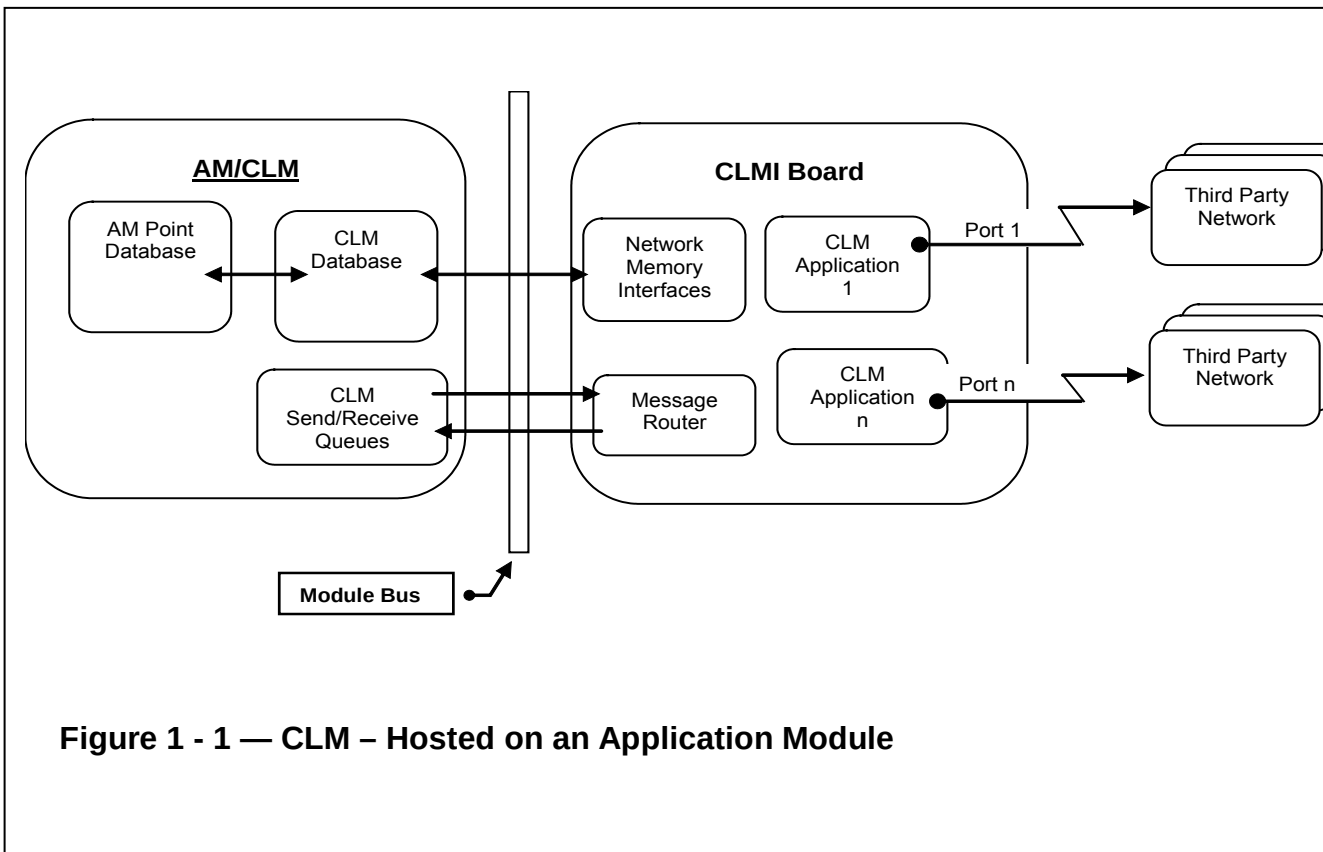
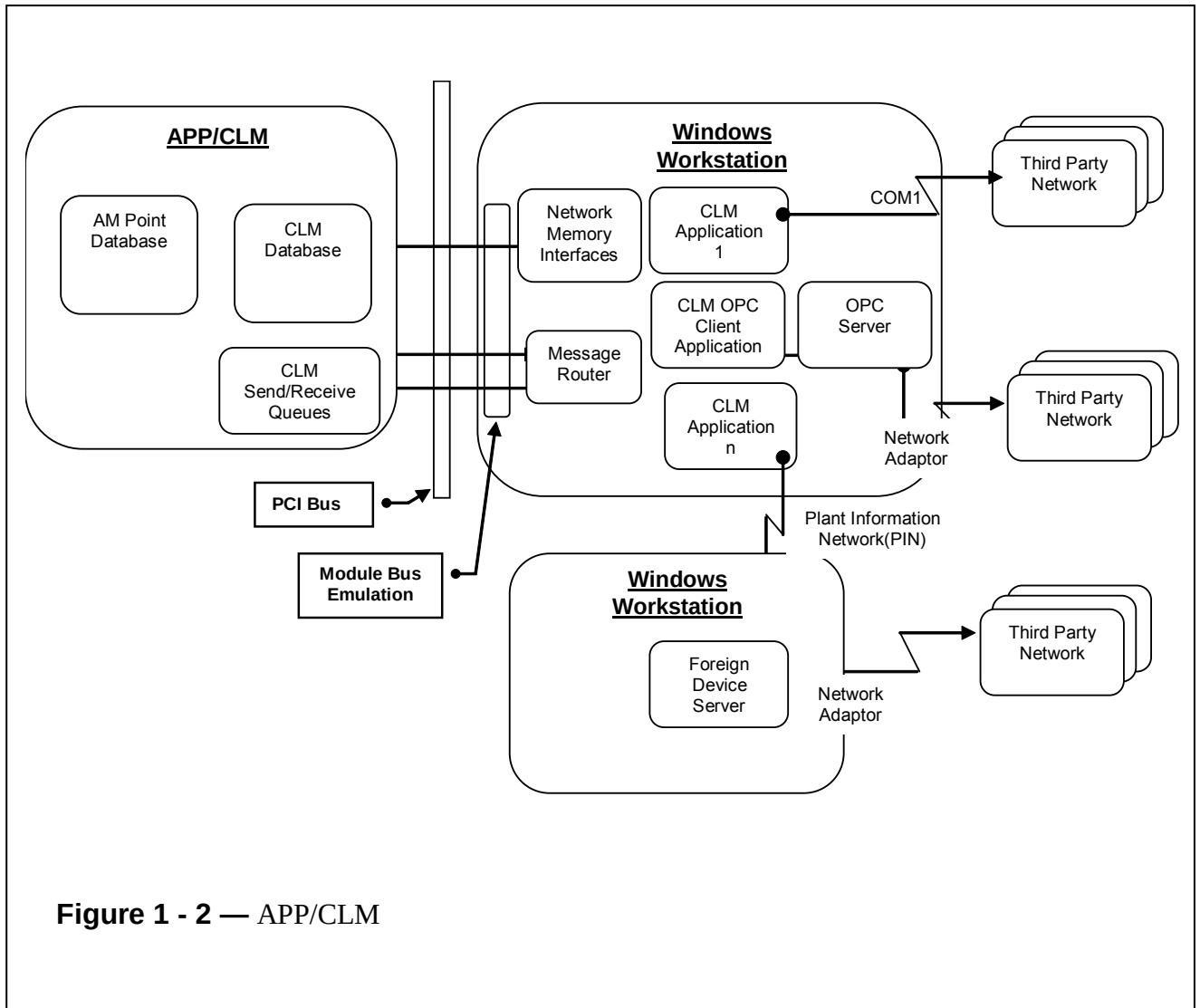


Figure 1 - 1 — CLM – Hosted on an Application Module

1.2.2. APP/CLM

The following figure depicts the architecture of the TPS Communications Link Module hosted on an APP.



1.3. Honeywell Support Services

Optional Installation Support, Field Services, and Technical Support are available during the on-site installation and checkout of TPS System equipment. Honeywell representatives are available to assist in interpreting this manual and to help resolve problems or situations not covered by this manual.

1.4. Related Publications

Honeywell's standard documentation is included with the system. The following publications apply to the TPS System and should be referred when required:

Title

TPS System Implementation Guide
APP Specification and Technical Data, TP03100B
Application Module Specification and Technical Data, AM03-500
Process Operations Manual
TPS/TPN Customer Release Guide
CLM Platform Planning, Installation LK02-100
TPS/CLM Platform Planning, Installation LK02-200 (this publication)
CLM Platform Implementation, LK12-340
CLM Parameter Reference Dictionary, LK09-240
CLM Process Operations, LK11-340

2. SITE PREPARATION

2.1. Site Requirements

Reference APP Installation Manuals, Sec. 1.4.

2.2. Configuration

Important! Redundant CLM configurations are **not** currently supported by the APP/CLM.

Up to two emulated CLMI boards may be configured (maximum of 4 communication ports per emulated board). Communications ports may be physical connections, such as COM ports on the APP, or logical ports, such as to an OPC Server. The APP/CLM normally comes with two COM ports available. An optional port expander (Stallion Card interface – HW model # **MZ-PCEB12**) may be used to increase the number of available COM ports to eight on the APP.

Depending upon certain communications limitations, the CLM physical connections can communicate either through modems or by a direct connection through the COM ports or Ethernet port. All cabling to CLM configured devices is through APP serial COM ports, additional communications boards or Ethernet communications.

Important! With the R60.12, CLM is supported on Windows 2003. The SYSTECH COM Port Extender needs to be used in place of Stallion Card.

2.2.1. Cabling Considerations

Certain restrictions and options must be considered in planning your installation.

2.2.1.1. Cable Lengths

The cables from the CLM COM ports cannot be longer than 15 cable-meters (50 cable-feet) when communicating RS-232. If the distance exceeds this limit, you must communicate RS-485 (maximum 1200 cable-meters or 4000 cable feet), use modems, or electrical-characteristics converters (ECC).

2.2.1.2. Direct Connection

If you are connecting a single device to one of the ports, and the cable length is less than 15 cable-meters, you may use an RS-232 direct connection (null modem) cable. In this arrangement, the RS-232 cable must be specifically wired to a connector that matches up with your device.

2.2.1.3. Typical cabling connections

Figure 2-1 illustrates a direct connect cable for connecting a APP/CLM 9-pin COM port to a 25-pin Modbus compatible device.

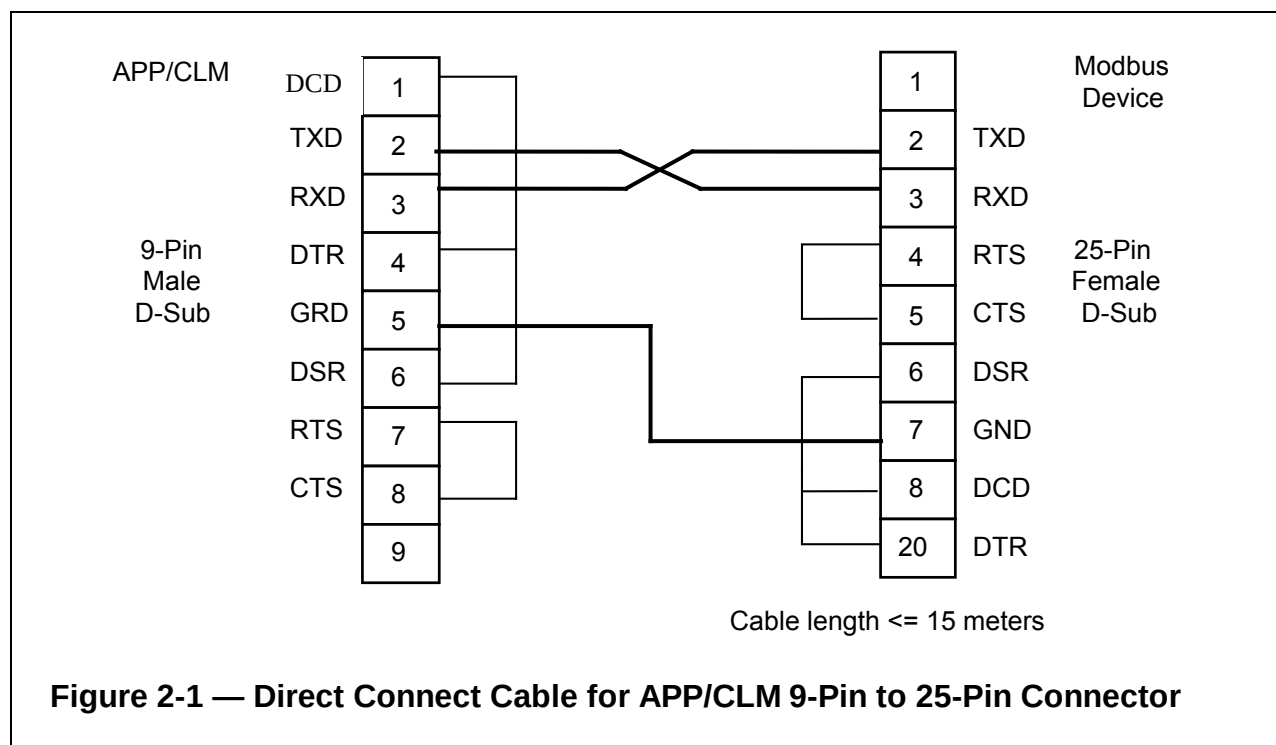


Figure 2-2 illustrates the modem cable for connecting a APP/CLM 9-pin COM port to a 25-pin modem or interface converter.

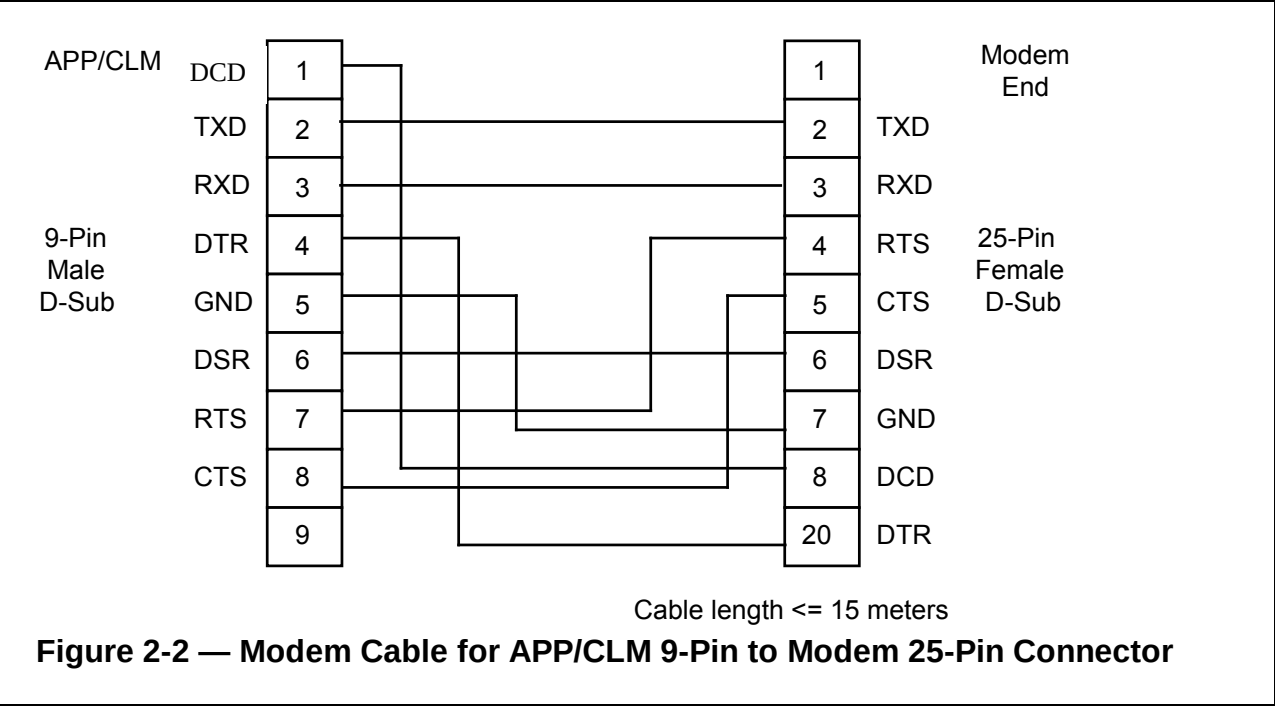
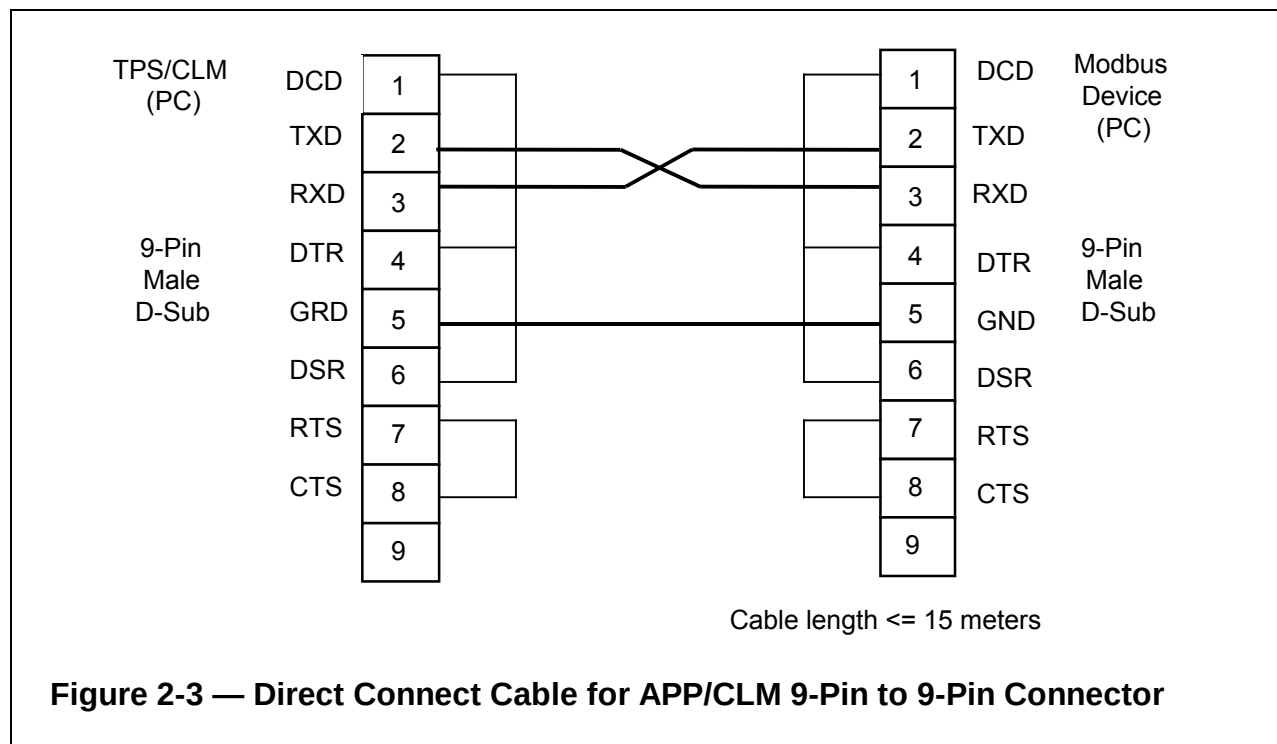


Figure 2-3 illustrates a direct connect cable for connecting a APP/CLM 9-pin PC COM port to a 9-pin Modbus compatible device PC COM port..



2.2.1.4. Cable connections for optional port expanders

If an optional port expander is used to utilize the maximum configurable ports, refer to the vendor documentation for the port expander for actual cable connections.

3. CLM Platform Software Installation

3.1. Introduction

3.1.1. APP-based Installation

There are two versions available for CLM/Platform Software installation.

- **CLM Platform** is the original CLM, which is installed to an AM node and requires the CLM communications boards, CLMI, CCP, or UCIM.
- **APP/CLM Platform** installs to the APP node with an AM personality installed and emulates the previous CLMI/CCP communications boards used in the earlier version of the CLM.

This manual describes **APP/CLM Platform** software installation to an APP. Refer to the *CLM Planning, Installation and Service Manual* for the AM-based CLM installation.

Important! With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32 bit and 64 bit variants).

3.2. APP/CLM Software – The Different Pieces

The LCN part of the CLM Platform software is added to the AM personality software on the Native (LCNP) side of the APP. This software is moved to the LCN from the CLM - LCN release media provided. The Native part of the APP/CLM appears very much like the AM-based CLM to the operator.

The TPS and the Experion part of the CLM Platform software is added to the Windows side of the APP using an installation script on the APP/CLM Platform Software release media provided.

CLM configuration rules are in the *CLM Platform Implementation Manual*.

Specific configuration and operating instructions as a CLM are in the *CLM Platform Implementation Manual*.

Standard AM Loading and operating procedures are in the *LCN Process Operations* binder.

3.2.1. APP/CLM Installation Overview

3.2.1.1. Installation overview for TPS APP

The APP/CLM is installed to the provided APP as follows:

- Determine the TPS node operating system (Windows 2000, 2003)
- Install the APP/CLM Platform software to Windows
- Install the APP/CLM Platform software (LCN);
 - Build CLM Platform displays
 - Add required AM support points
 - Include required information for the node in the NCF (Modify)
- Install the CLM application software, such as Modbus.

Important! The users have the option to install and configure the required protocol. Reference the specific application manual for the respective installation procedure.

The following is required to complete an installation with an application and is specific to each application – the process is presented here as an overview.

To add an application:

1. Establish the physical, cable, or logical port connection
2. Install the APP/CLM application per the application's manual
3. Configure the application
4. Build the application in the network devices
5. Define and build the CLM image points to image the network data
6. Build the application's point interface table structures to connect the network device points to the CLM image points
7. Verify application communications and operations

3.2.1.2. Installation overview for Experion APP

With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32 bit and 64 bit variants).

The APP/CLM is installed to the provided APP as follows:

- Determine the Experion node operating system (Windows 2008).
- Install the APP/CLM Platform software to Windows.
- Install the APP/CLM Platform software (LCN);
 - Build CLM Platform displays.
 - Add required AM support points.
 - Include required information for the node in the NCF (Modify).
- Install the CLM application software, such as Modbus.

Important! The users have the option to install and configure the required protocol. Reference the specific application manual for the respective installation procedure.
--

The following is required to complete an installation with an application and is specific to each application – the process is presented here as an overview.

To add an application:

1. Establish the physical, cable, or logical port connection.
2. Install the APP/CLM application per the application's manual.
3. Configure the application.
4. Build the application in the network devices.
5. Define and build the CLM image points to image the network data.
6. Build the application's point interface table structures to connect the network device points to the CLM image points.
7. Verify application communications and operations.

3.3. Windows 2003 Only - Install the TPS-Based Software to the APP/CLM

Important! This section is only applicable for users who will continue to use the TPS node operating system (Windows 2003).

3.3.1. Windows 2003 Only - Shutdown TDC Emulation Service

3.3.1.1. Shutdown TDC Emulators Service

In the Windows environment, the Honeywell TDC Emulators Service simulates the peripheral boards that exist in a traditional TPN node chassis. In the APP/CLM, this service simulates the CLMI and CCP boards found in a traditional CLM. Before installation of APP/CLM software, this service should be stopped.

Note! The APP/CLM installation is now supported in Windows 2000. The installation steps remain same for Windows 2000.

Step	Action
1	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	From the Start menu select Settings/Control Panel/Administrative Tools.
3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 3.3-1 below.
4	If the Status for TDC Emulators Service is Started , select the TDC Emulators Service entry. From the drop-down menu, select Action , then the Stop button.
5	A message box appears indicating that the service is being stopped. This will take about 30 seconds. See Figure 3.3-2 below. Note: Occasionally, the TDC Emulators Service does not stop in the required time and an Error Message box appears with the message <i>“Could not stop TDC Emulators Service service”</i> . When this happens, press the OK button in the message box. The TDC Emulators Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt and Del keys simultaneously and invoking the Task Manager. From the Task Manager, view the Process tab and verify that the tdcaserver.exe process is no longer in the active process list.

The following figures illustrate the TDC Emulators Service.

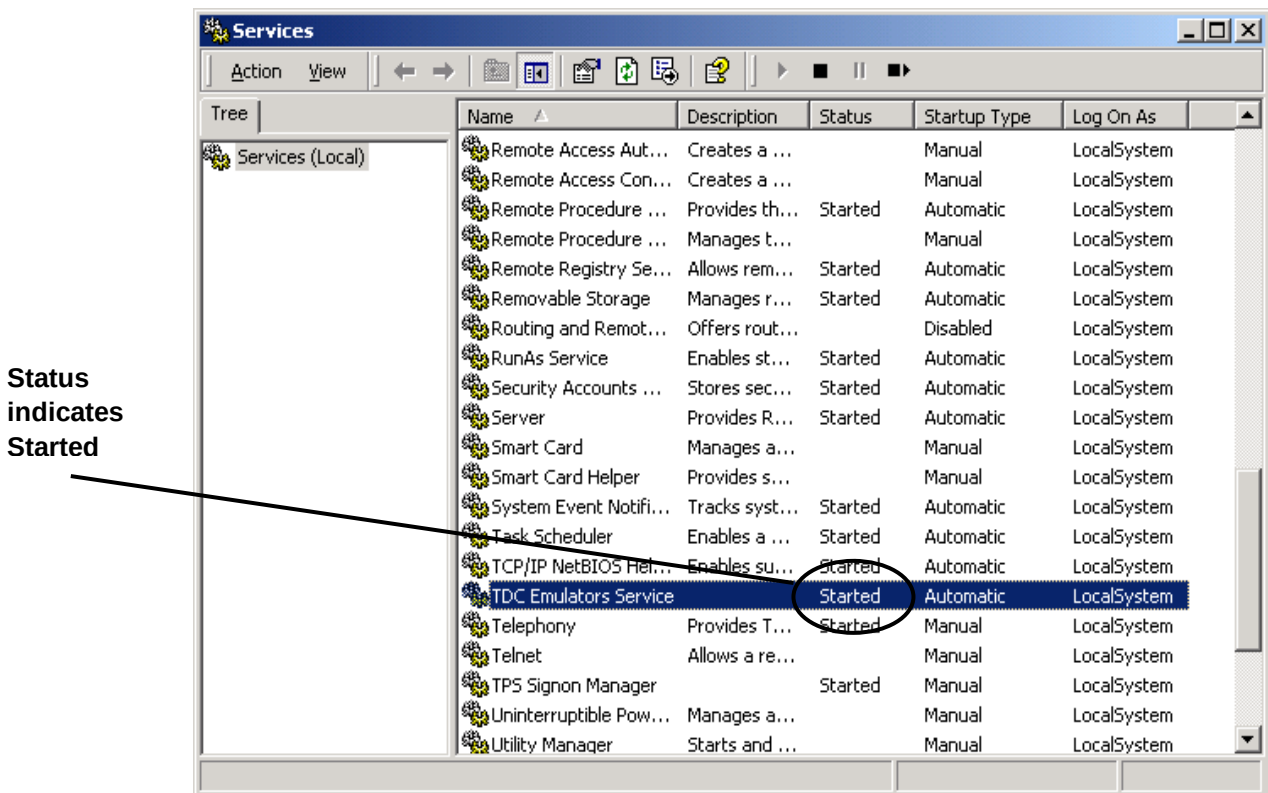


Figure 3.3-1 TDC Emulators Service – Started/Running

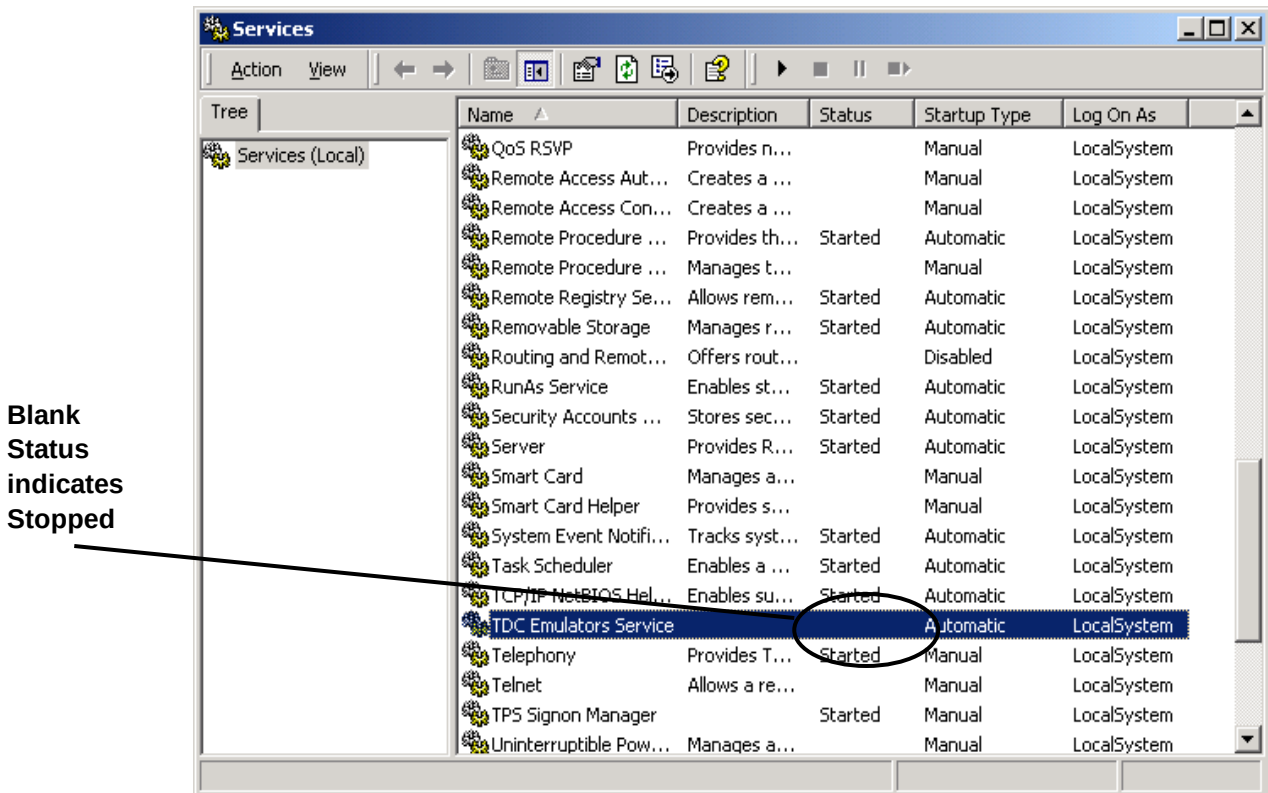


Figure 3.3-2 TDC Emulators Service – Stopped

3.3.2. Windows 2003 Only - Install TPS/CLM Windows 2003-Based Software

3.3.2.1. Install the Windows 2003-based Software

To install the APP/CLM Platform software on the TPS APP, **do** the following steps.

Step	Action
1	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	Insert the APP/CLM Platform Software distribution media into a drive.
3	View the distribution media in the drive using Windows Explorer. From Windows Explorer, launch the CLM Platform Install.exe program from the \TPS2K directory on the distribution media.
4	Follow the instructions for installation. It is recommended that all the defaults be selected.
5	Confirm that the installation is completed successfully.

3.3.3. Windows 2003 Only - Configure the Emulated Board(s)

3.3.3.1. Configure the Logical Boards

Configure the APP/CLM emulated interface board(s) on the TPS APP, using the Configuration Utility.

Step	Action
1	From the Start menu select <i>Programs/Honeywell TPS</i> and start the <i>Configuration Utility</i> .
2	From the <i>Configure</i> menu select <i>Virtual Board 1</i> . The CLM Configuration page will appear as shown in Figure 3.3-3.
3	Enable the Check Box for CLMI Interface. From the drop down menu, Select +Warning (EW) Select the radio button corresponding to slot to be configured – this should normally be slot 2 or 3 (slot 4 can have other uses). This slot selection must match the slot defined later in CLMI0nn.ZC file – these two items must match. Note: CLMI0nn.ZC is described in 3.3.5 - Configure the CLMI.
4	Select the Apply button and exit the TPS Configuration Utility.

5	Repeat if required for a second CLMI configuration on Virtual Board 2.
---	---

The following figure illustrates the CLM Configuration page.

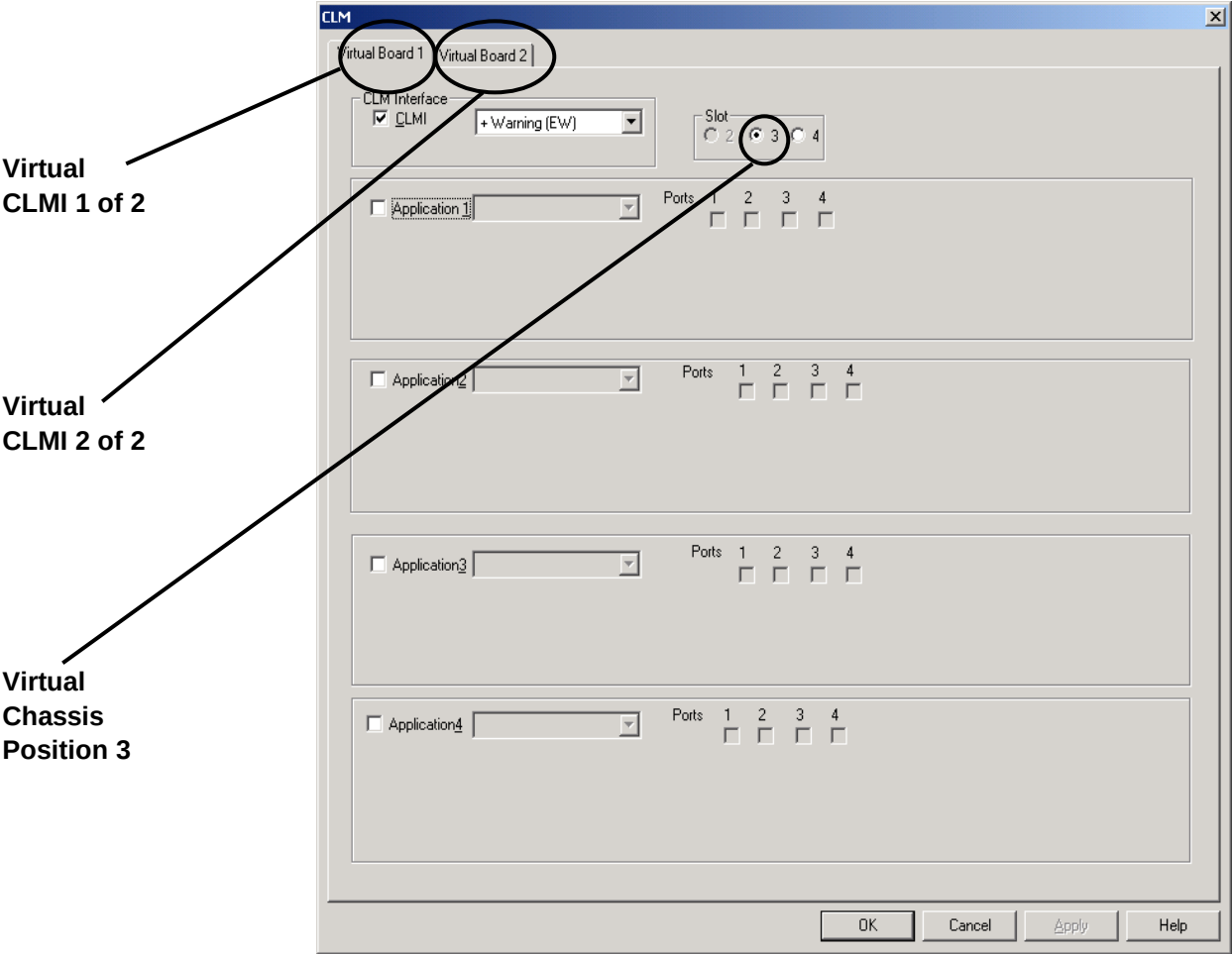


Figure 3.3-3 CLM Configuration Page

3.3.4. Windows 2003 Only - Start TDC Emulation Service

3.3.4.1. Start TDC Emulators Service

The TDC Emulators Service simulates the CLMI and CCP boards found in a traditional CLM. Before the TPN personality can be loaded, the TDC Emulators Service must be started and running.

Important! Anytime the CLM configuration is changed from the CLM page of the TPS Configuration Utility, the TDC Emulators Service must be stopped and re-started before the changes will take effect.

Step	Action
1	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	From the Start menu select Settings/Control Panel/Administrative Tools.
3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 3.3-2 previously. The current status should be Stopped.
4	If the Status for TDC Emulators Service is Stopped, press the Start button. When a box appears for confirmation, press Yes. If the Status for TDC Emulators Service is not Stopped, stop the service as described above and then re-start it. See 3.3-1 previously. Note: If the Startup state of the service is Manual, the state must be changed to Automatic so that the service will start itself on restart or initial boot of the APP. To make this change, do the following: • From the drop-down menu, select Action\Properties. • From the drop-down menu, select Automatic. • Select OK to save the changes. See Figure 3.3-4.
5	At this point, APP/CLM-based applications may be installed to the APP/CLM. The TPS node be reset from the LCNP Status Display before loading the application. To accomplish this, from the Start menu, select Programs/Honeywell TPS/ LCNP Status, and press the Reset button. See Figure 3.3-5.

The following figures illustrate the TDC Emulators Service.

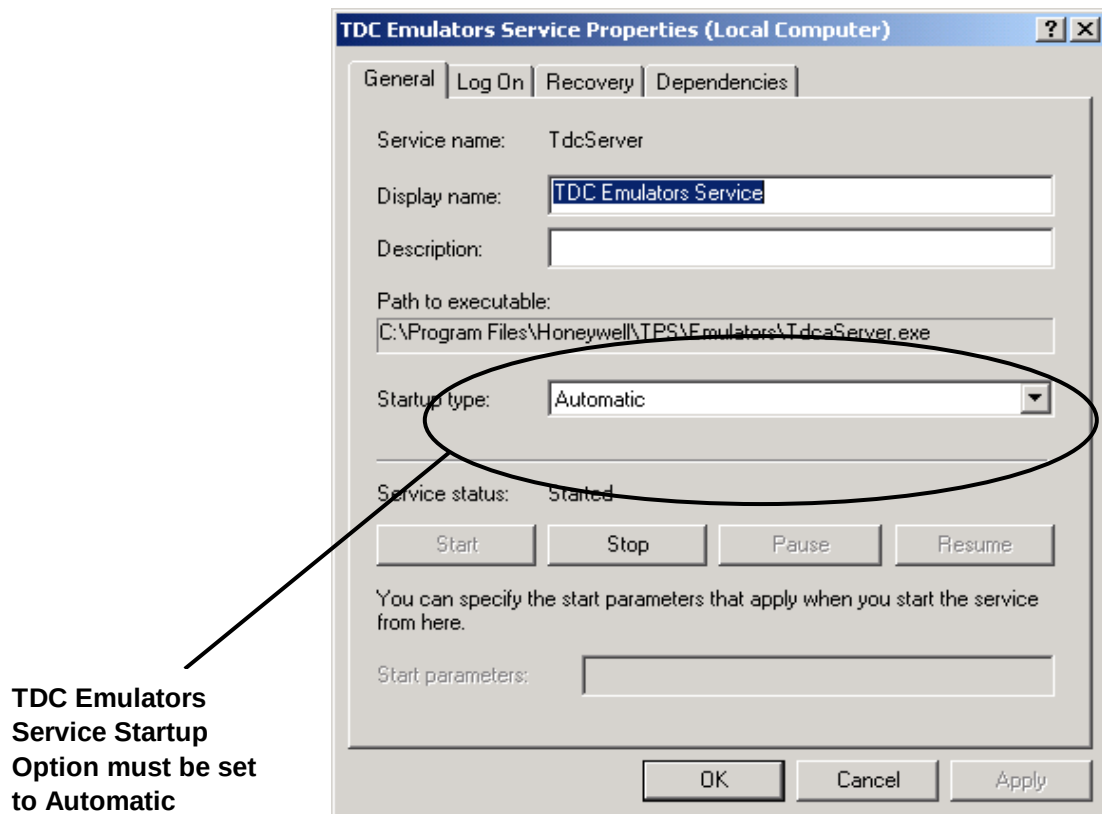


Figure 3.3-4 TDC Emulators Service – Change to Automatic Startup

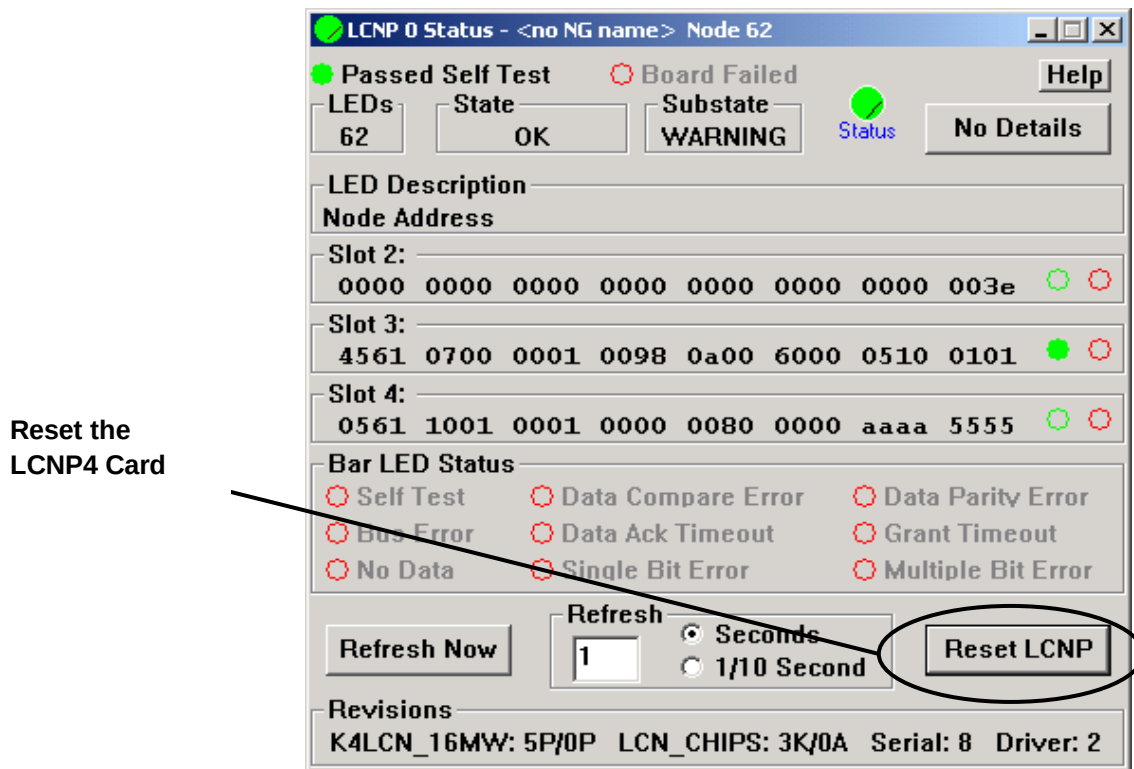


Figure 3.3-5 LCNP Status Display – Reset the Node

3.3.4.2. The next step

This concludes the installation of Windows 2003-based software to the **APP/CLM**.

The next step is to install the LCN-based software to the AM (Native) side of the **APP/CLM**.

Proceed to Section 3.5.

3.4. Windows 2008 Only - Install the Experion-Based Software to Experion APP

Important! This section is only applicable for users who will be using the Experion node operating system (Windows 2008 platform on both 32 bit and 64 bit variants).

3.4.1. Windows 2008 Only - Shutdown TDC Emulation Service

3.4.1.1. Shutdown TDC Emulators Service

In the Windows environment, the Honeywell TDC Emulators Service simulates the peripheral boards that exist in an Experion node chassis. In the APP/CLM, this service simulates the CLMI and CCP boards found in a traditional CLM. Before installation of APP/CLM software, stop this service.

Step	Action
1	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	From the Start menu select Control Panel/System and Security/ Administrative Tools.
3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 3.4-1 below.
4	If the Status for TDC Emulators Service is Started , select the TDC Emulators Service entry. From the drop-down menu, select Action, then the Stop button.
5	A message box appears indicating that the service is being stopped. This will take about 30 seconds. See Figure 3.4-2 below. Note: Occasionally, the TDC Emulators Service does not stop in the required time and an Error Message box appears with the message “Could not stop TDC Emulators Service service”. When this happens, press the OK button in the message box. The TDC Emulators Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt and Del keys simultaneously and invoking the Task Manager. From the Task Manager, view the Process tab and verify that the tdcserver.exe process is no longer in the active process list.

The following figures illustrate the TDC Emulators Service.

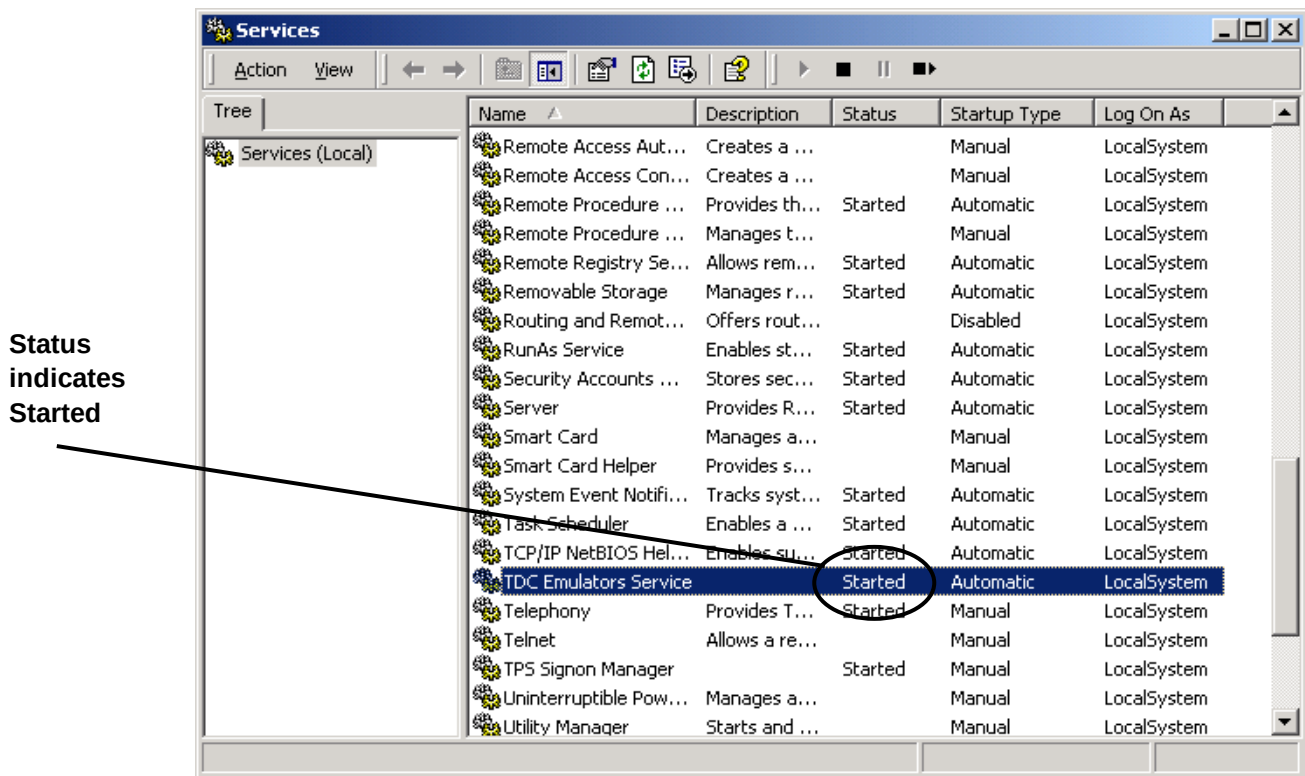


Figure 3.4-1 TDC Emulators Service – Started/Running

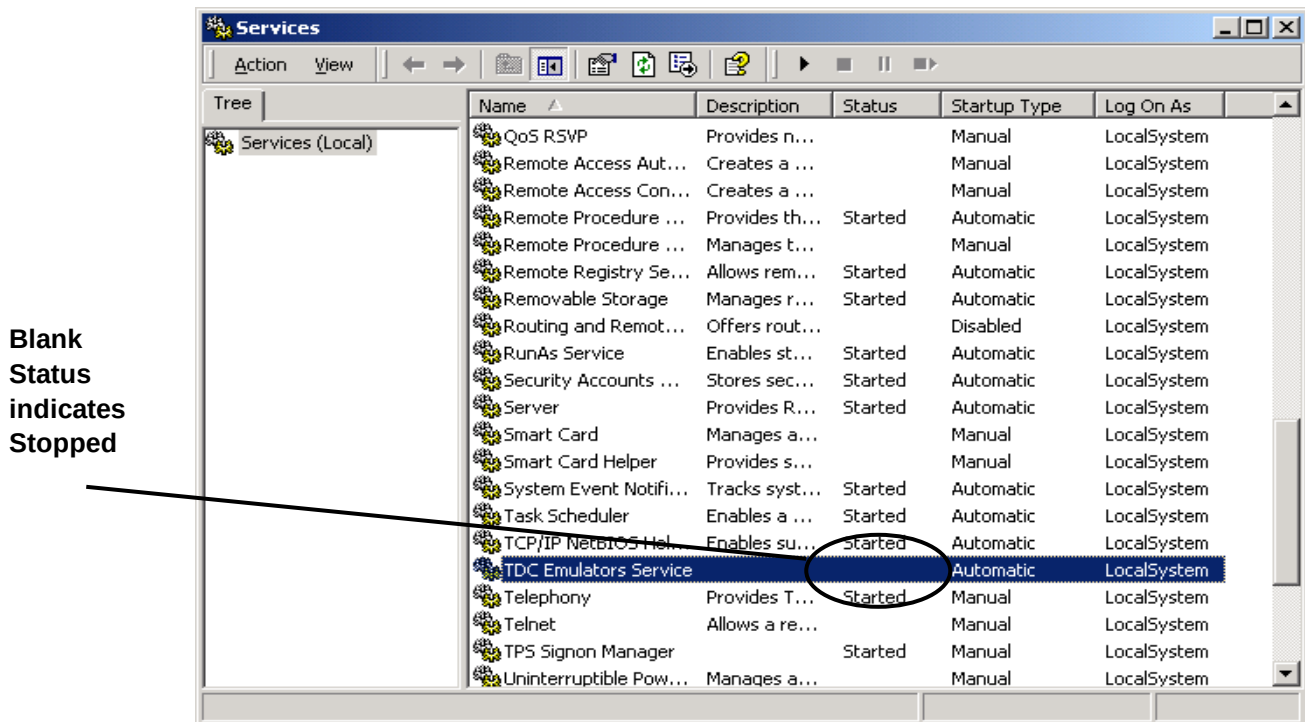


Figure 3.4-2 TDC Emulators Service – Stopped

3.4.2. Windows 2008 Only - Install APP/CLM Windows 2008-Based Software

3.4.2.1. Install the Windows 2008-based Software

To install the APP/CLM Platform software on the Experion APP, **do** the following steps.

Step	Action
1	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	Insert the APP/CLM Platform Software distribution media into a drive.
3	View the distribution media in the drive using Windows Explorer. From the Windows Explorer, launch the CLM_Platform_Install.msi program by double clicking from the \TPSCLM200 directory on the distribution media.
4	Follow the on-screen instructions for installation. It is recommended that all the defaults be selected.
5	Confirm that the installation is completed successfully.

3.4.3. Windows 2008 Only - Configure the Emulated Board(s)

3.4.3.1. Configure the Logical Boards

Configure the APP/CLM emulated interface board(s) on the Experion APP, using the Configuration Utility.

Step	Action
1	From the Start menu select Programs/Honeywell Experion PKS/System Management and start the Configuration Utility .
2	From the Configure menu select CLM. A dialog box appears.
2	Select Virtual Board 1 . The CLM Configuration page will appear as shown in Figure 3.4-3.
3	Enable the Check Box for CLMI Interface. From the drop down menu, Select +Warning (EW) Select the radio button corresponding to slot to be configured – this should normally be slot 2 or 3 (slot 4 can have other uses). This slot selection must match the slot defined later in CLMI0nn.ZC file – these two items must match . Note: CLMI0nn.ZC is described in 3.4.5 - Configure the CLMI.
4	Select the Apply button and exit the Experion Configuration Utility.
5	Repeat if required for a second CLMI configuration on Virtual Board 2 .

The following figure illustrates the CLM Configuration page.

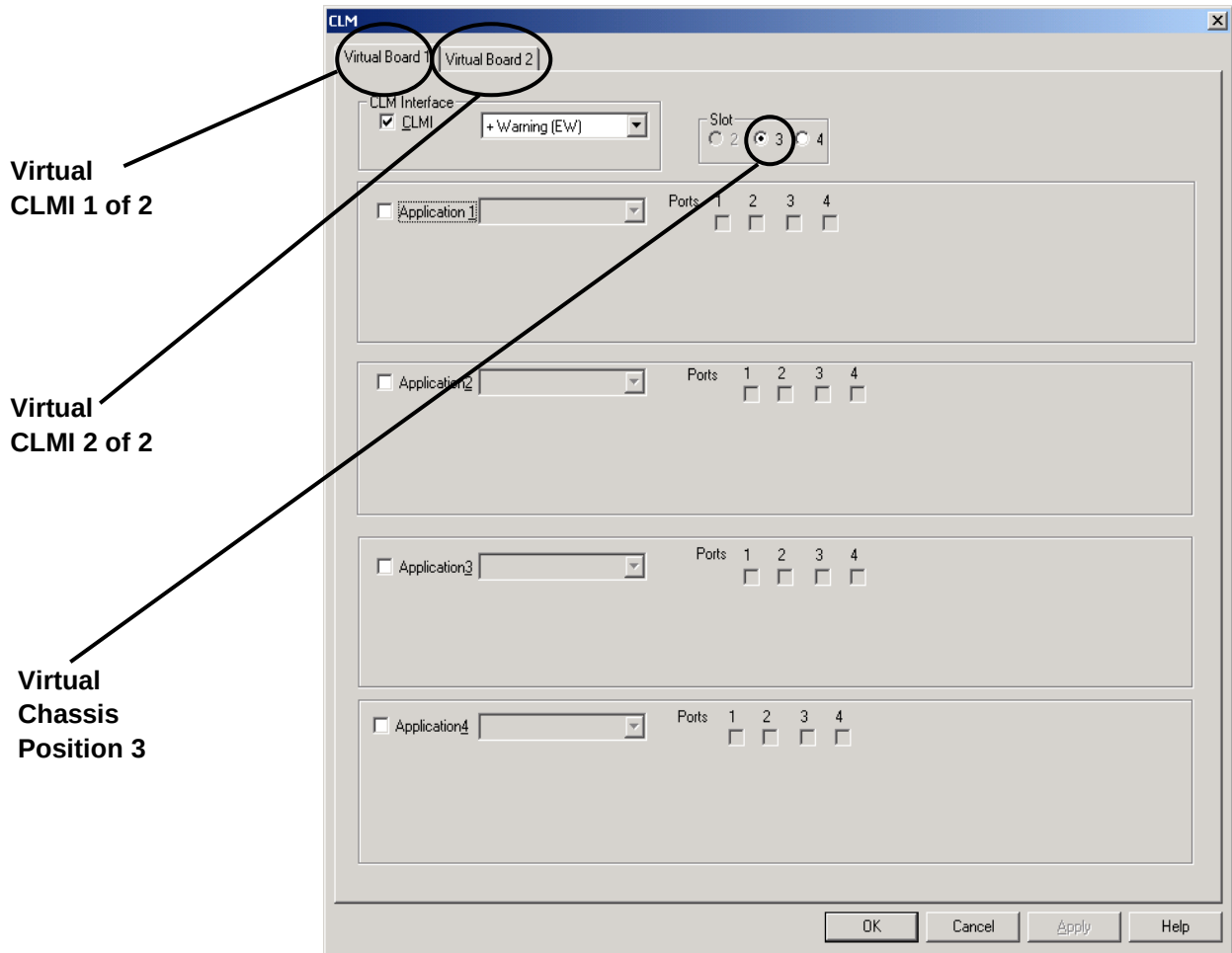


Figure 3.4-3 CLM Configuration Page

3.4.4. Windows 2008 Only - Start TDC Emulation Service

3.4.4.1. Start TDC Emulators Service

The TDC Emulators Service simulates the CLMI and CCP boards found in a traditional CLM. Before the TPN personality can be loaded, the TDC Emulators Service must be started and running.

Important! Anytime the CLM configuration is changed from the CLM page of the Experion Configuration Utility, the TDC Emulators Service must be stopped and re-started before the changes will take effect.

Step	Action
1	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2	From the Start menu select Control Panel/System and Security/ Administrative Tools.
3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 3.4-2 previously. The current status should be Stopped.
4	If the Status for TDC Emulators Service is Stopped, press the Start button. When a box appears for confirmation, press Yes. If the Status for TDC Emulators Service is not Stopped, stop the service as described above and then re-start it. See 3.4-1 previously. Note: If the Startup state of the service is Manual, the state must be changed to Automatic so that the service will start itself on restart or initial boot of the APP. To make this change, do the following: • From the drop-down menu, select Action\Properties. • From the drop-down menu, select Automatic. • Select OK to save the changes. See Figure 3.4-4.
5	At this point, APP/CLM -based applications is installed to the APP/CLM. The Experion node should be reset from the LCNP Status Display before loading the application. To accomplish this, from the Start menu, select Programs/Honeywell Experion PKS/ LCNP Status, and press the Reset button. See Figure 3.4-5.

Important!

- In 32-bit Windows 2008 based Experion APP, tdcaserver.exe (TDC Emulators Service) is located under C:\Program Files\Honeywell\TPS\Emulators\tdcaserver.exe.
- In 64-bit Windows 2008 based Experion APP, tdcaserver.exe (TDC Emulators Service) is located under C:\Program Files (x86)\Honeywell\TPS\Emulators\tdcaserver.exe.

The following figures illustrate the TDC Emulators Service.

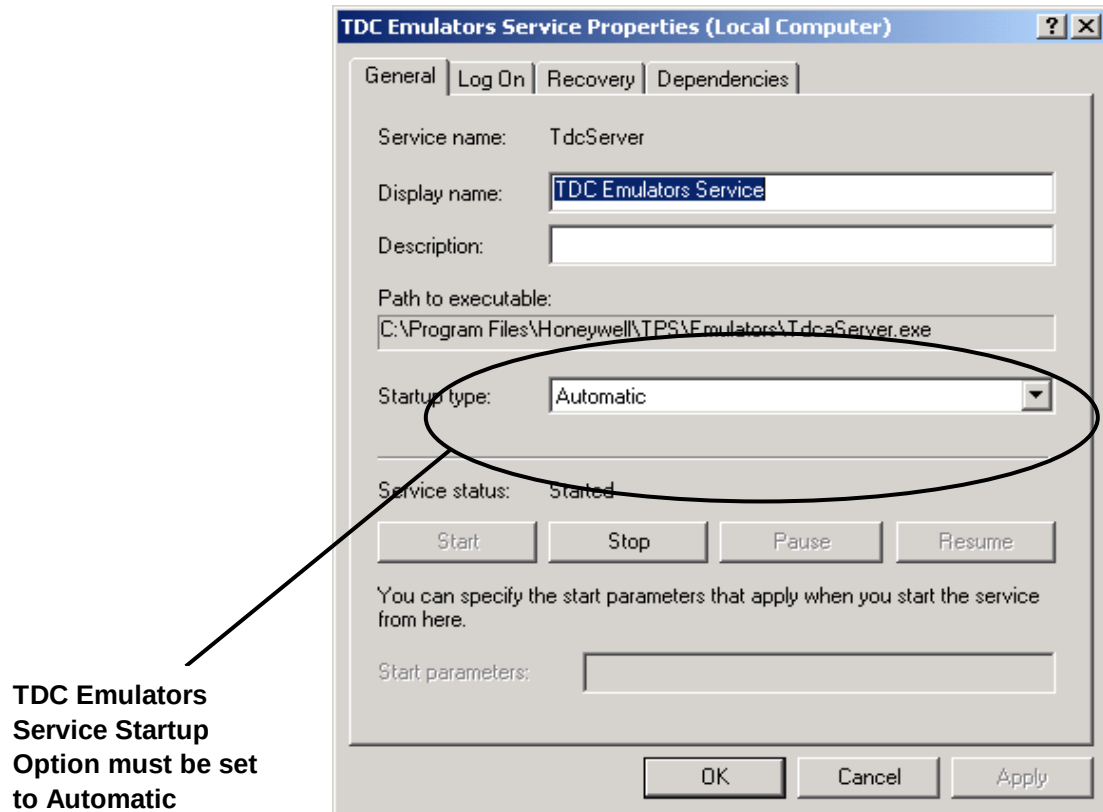


Figure 3.4-4 TDC Emulators Service – Change to Automatic Startup

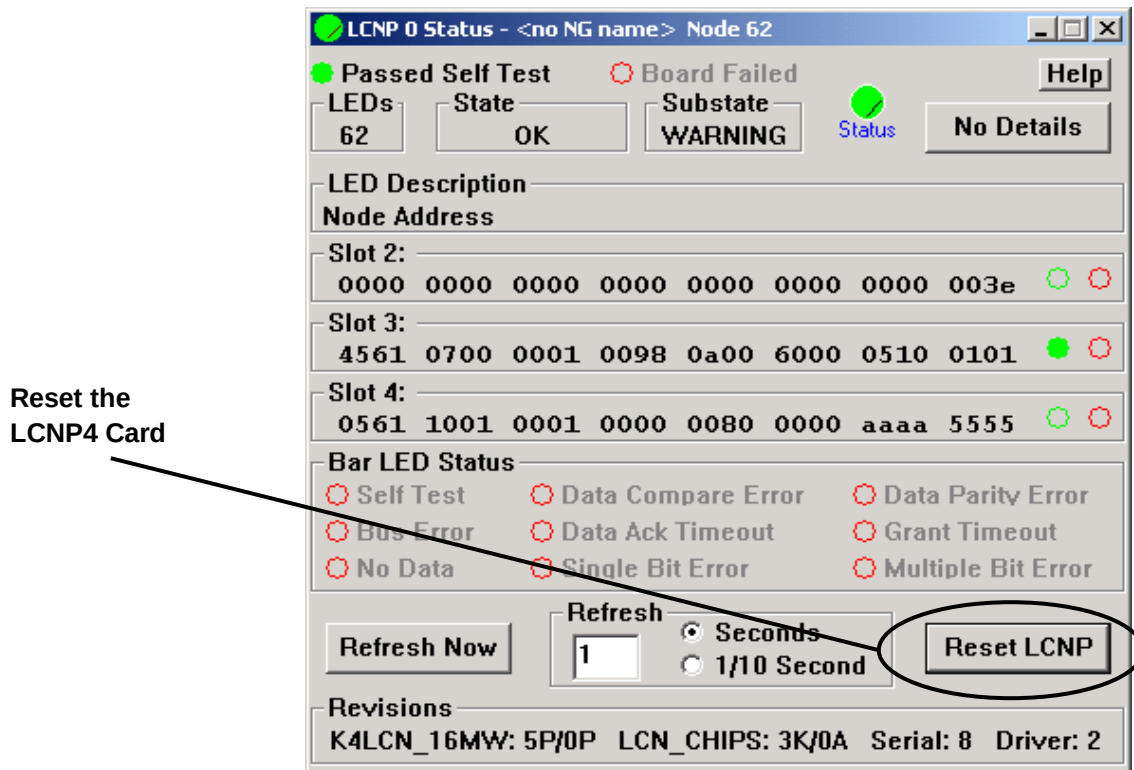


Figure 3.4-5 LCNP Status Display – Reset the Node

3.4.4.2. The next step

This concludes the installation of Windows 2008-based software to the **APP/CLM**.

The next step is to install the LCN-based software to the AM (Native) side of the **APP/CLM**.

Proceed to Section 3.5.

3.5. Install the LCN-Based Software to the APP/CLM

The following section describes the procedures for the installation of the CLM Platform LCN-based software.

3.5.1. Overview

3.5.1.1. Configuration overview

The **APP/CLM** is configured in an APP node by:

- Adding CLM Platform software
- Building required LCN support points
- Building CLM operating displays
- Adding required information for the node in the NCF (Modify)
- Adding one or more applications by
 - establishing the physical port connections
 - configuring the non-TPN network
 - building the application in the non-Honeywell devices
 - configuring AM-based points to image the non-Honeywell controller points
 - building the point interface table structures to connect the non-Honeywell controller points to these image points

The following subsections describe the procedures required to accomplish the platform installation outlined above.

Before performing the platform installation, you need to understand the implication of *unit names*. This is described in Section 14 of the *Engineer's Reference Manual*, located in the *Implementation/Startup and Reconfiguration - 2* binder.

3.5.2. Install CLM Platform Software

3.5.2.1. Before You Do the Software Installation

1) Create a directory in one of the destination user volumes.

- The directory name of “UCLM” is suggested and used in this procedure.
- This directory must have a minimum of 15 megabytes of disk space available and a maximum file setting of at least 2000.

2) From the **Modify Volume Paths** display, set the path for:

- CL SOURCE/OBJ to NET>UCLM>
- CL PARAM LIST* to NET>UCLM>
- USER DEFLT PATH* to NET>UCLM>
- *(same path as the CL SOURCE/OBJ).
- CL CUSTOM GDF to NET>CDSG>
- (verify the directory CDSG exists)

3) From the Command Processor, verify that:

- &CUS and &CLX directories also exist (this was completed during the installation of the Application Module software)
- Maximum Files is at least 2000 and free disk space is at least 15 mbytes.

Note:

Some files included in the product release are copied to your system wherever specified within the install process.

Source files for custom displays are included and should be put into directories included in the schematic search rule (UCLM is suggested.).

The following table is a summary of the CLM Platform LCN software installation.

Step	Description
1	Execute the provided install EC file and check for errors.
2	Build a CLM Network Status Point and a Builder Point for each CLM Node installed on the system.
3	Customize and compile CLM Platform operating displays.
4	Define the configuration required for communications.

3.5.3. Execute the Install Procedure

In preparation for executing the install procedure:

- Read the TPS/TPN Customer Release Guide, for information that applies to your installation including HM sizing requirements, special instructions, and reinstallation information.
- Mount the CLM Platform release media.
- Visually scan the file - INSTALL.EC - provided on the supplied media, for an overview of the install process.

Important – review the CDS packages to be installed and compiled to assure **no conflict** with existing CDSs on the system, that is, no duplicate names.

The install EC file performs the following functions:

- checks for previous installation of the package.
- prompts for the current TPS release for loading of the specific release dependent files to the HM. Some files to be loaded are independent of the LCN release. Other files must be loaded or compiled only for a specific release.
- prompts for confirmation of load path (directories) setup.
- copies required files from the supplied media to the designated HM directories.
- compiles supplied CDSs required for package operation.
- creates a file, INSTALL.LS, of the install execution for verification of correct installation, no errors listed, and for future reference.

Do the following 3 steps to complete the EC installation:

STEP 1:
Initiate the EC File

In the Command function, within the Engineering Personality, enter the following command to begin the install procedure:

EC \$Fn>&CLM>INSTALL.EC \$Fn

where n = the logical device identification where the CD is mounted (for example, \$F1).

STEP 2:
Follow the Prompts

Follow the prompts output by the install execution and enter the required information to continue through the procedure.

Caution: An incorrectly formed pathname entry may cause the INSTALL script to abort prior to completion. If this occurs, the INSTALL script may be restarted by setting the prompt data off (type PD OFF and press ENTER) and closing the verification output file (type DO and press ENTER), then re-executing the INSTALL.EC.

STEP 3:
Verify the Results

Print out the INSTALL.LS file and confirm that there were no errors during the execution of the EC file.

3.5.4. Build Required CLM Platform Points

3.5.4.1. Build the Required CLM_nn & BUILD_nn LCN Points

A CLM Network Status Point and a Builder Point are required for each CLM node configured. These points provide alarming and display support to the CLM Platform. These points can be generated by modifying the EB files included in the release as explained below.

From the Command Processor, **do** the following steps

Step	Action
1	Set the current pathname to NET>UCLM>.
2	- Copy the CLM_NN.EB file to another .EB file following CLM naming convention where NN is the appropriate CLM Node number (e.g. CLM_55.EB). - Edit the new file, changing the point name, unit, and optionally the description, and keyword to the desired values. Notes: The processing cycle (period) has a default value of 30 seconds, but can also be changed. The CDS packages attached to the point are NETCDS, NETCDS1, NETCDS2, NETCDS3, NETCDS4 These were compiled in the execution of the INSTALL.EC task. They are required for the CLM_NN point.
3	- Select the BUILDER COMMANDS from the Engineering personality menu. - Select EXCEPTION BUILD - Select the Load Entities option with your .EB file identified as the input using CLM_NN as the IDF file name. - Initiate an exception build of the point into the CLM/AM database.
4	- Copy the BUILD_NN.EB file to another .EB file following the CLM naming convention where the NN is the CLM node number (e.g., BUILD_55). - Edit the new file, changing the point name, unit, and optionally the description and keyword to the desired values. The CDS packages attached to the point are BUILDCDS, BLD2CDS, CLMDCDS These CDSs were compiled in the execution of the INSTALL.EC task. They are required for the BUILD_NN point.

Step	Action
5	• Select the Builder Commands from the Engineering personality menu. Select EXCEPTION BUILD • Select the Load Entities option with your .EB file identified as the input using BUILD_NN as the IDF file name. • Initiate an exception build of the point into the TPN database.
6	• Call the Detail display of the new BUILD_NN point. Set the initial default pathnames by entering appropriate text into the following string parameters. • Move to the Custom pages and change the following parameters. ENTARR(101) = &CUS <Default = CLM Save Data > ENTARR(102) = NET>&CUS <Default = CLM Builder Support > ENTARR(103) = NET>&CUS <Default = CLM Application > Note: After the CLM custom displays are compiled in the next step, these pathnames may be changed or viewed from the CLM CONFIG DATA page.
7	• Now checkpoint the CLM: - Go to the Application Modules Status Display, - Select the CLM/AM node, and - Select SAVE DATA to checkpoint to the HM (default destination).
8	Repeat Steps 2 through 7 for each additional CLM.

3.5.5. Customize Displays

3.5.5.1. Custom Graphics Must be Compiled

The custom displays required by the CLM Platform are included in the release media and were copied to the HM in the previously executed INSTALL.EC process.

The CLM Status Display "CLMSTAT1" must be modified to include the CLM node(s). If other CLMs are added to the LCN later, they must be added to the CLM Status Display by following this same procedure.

Do the following steps to customize the displays.

Step	Action
1	From the Engineering Menu start the Command Processor. Set the current path to the directory where the CLM displays are stored (usually <i>NET>UCLM</i>) by entering: SP NET>UCLM>
2	Invoke the Picture Editor by entering the command: PE
3	Load the Global Database names by entering the command: LD CLMDDB
4	Read in the display by entering the command: R CLMSTAT1
5	Modify the display by entering the command: SEL VAR;MOD;
6	At the top of the display are 20 sub-pictures in two columns indicated by the text: <i>SUB CLMID1</i> Select the first sub-picture text in the left column. Then press <i>ENTER</i> .
7	Make the following change in the variant text. Change: IF 1 <> 1 THEN "SUB CLMID1" To read: IF 1 = 1 THEN SUB CLMID1 NOTE: Remember to remove the “ “ marks. Then press <i>ENTER</i> .
8	A screen appears that allows you to enter information about a CLM: ENTER THE LCN CLM POINT ID Enter the name of the CLM point built previously (CLM_nn). ENTER Node Number (e.g. 49) Enter the LCN node number (01-96). ENTER Redundant Node Number If the CLM is redundant, enter the backup node number (nn+1)
9	Repeat steps 5 through 8 for each CLM to be added to the display.

Step	Action
10	Compile the display by entering the command: COM Press ENTER to overwrite (save).
11	All of the other custom displays included with the CLM must now be compiled. The Picture Editor Multiple Compile (MCOMP) command is used to compile all required displays. Enter: LD CLMDDB MCOMP NET>UCLM>CLM_DS Check the CLM_DS.ER file for compilation results.
12	Exit the Picture Editor by entering: END

ATTENTION

Important Notes!

The schematic "CLMSCHEM" is the initial display in the CLM Network Display process. Its function is to establish the required values in the Honeywell variable area.

When this display is read in and compiled, *it will look like a blank screen*. In fact, it does have a process included in the "Initial" target attached to the display.

The schematic "CLMSCHEM" is the lead display and can be put on a button, if desired. It can also be accessed using the "SCHEM" key or from the "VIEW OBJECT DETAIL" target of one of the CLM objects on the status detail display. Refer to the *CLM Process Operations Manual*.

Update the area database with the pathname to the graphics, if required.

Be sure that the unit with the CLM_nn and BUILD_nn points is assigned on the console where the CLM displays are used.

3.5.6. Configure the Emulated CLMI Board

3.5.6.1. You Must Define the Emulated Board Location

This step defines the configuration information required by the platform to identify the emulated board slot location and logical board number (1-2). This information is entered as if the physical AM chassis existed and the CLM interface board was inserted into one of the physical slots.

Part of this configuration information resides in a specific file for each different CLM node (or redundant node pair) on the LCN (sample file CLMI nnn .ZC). The other part is done through the Configuration Utility.

The CLMI configuration file is an ASCII text file with the following characteristics:

The CLMI nnn .ZC File

- CLMI nnn .ZC
where nnn = CLM physical node number (000-096) or the lower of the node numbers of the redundant pair.
- Located in the directory of NET>&CUS> by default or the configured path indicated on the CLM Configuration page.

ZC File Directives

- The following are the supported configuration directives:
CLMI <board #>, <slot #>
END
- One set of operating system directives (CLMI and END) are required for each CLM Interface board configured.
- The directives supported on the **APP/CLM** Loader Directive File are a subset of those supported on the AM-based CLM. The application specific directives are not required by the **APP/CLM** since that information is entered through the Configuration Utility.

3.5.7. Configure the Emulated CLMI Board, continued

3.5.7.1. Edit the Configuration File

- Copy CLMI_{nnn}.ZC to a new file, replacing nnn with the actual three digit node number.

Example file name: NET>&CUS>CLMI024.ZC

- Invoke the LCN Editor and read in the new ZC file.
- Determine and enter the logical board number and physical chassis slot location.

Important!	The information entered into the CLMI directive statement in the configuration file <u>MUST</u> match exactly the information entered into the CLM Configuration Utility process.
-------------------	--

- Repeat for a second board, if applicable.
- Save the file and exit.

The following is an example of a configuration file:

*

* EXAMPLE CONFIGURATION FILE (.ZC)

*

* FILE NAME: NET>&CUS>CLMI024.ZC

*

CLMI 1,3 (Board #1 in slot #3)

END

Note 1: The slot # specified here must be the same # referenced in the CLM Windows Configuration Utility – Section 3.2.3. This should be 2 or 3.

Note 2: Earlier CLM Platform packages defined port application loading directives (LOAD, RUN, and INIT) in the ZC file. These directives are not used for the **APP/CLM** and this function has moved to the CLM application installation instructions for each application. See the application manual.

3.5.8. Update LCN Network Configuration File

3.5.8.1. NCF update process

The LCN Network Configuration File (NCF) must be updated to identify each AM that is configured as a CLM.

The two steps for safely updating the LCN Network Configuration File (NCF) are:

Step	Description
1	Setting up for backup of the existing NCF to removable media.
2	Updating the NCF for each new CLM and error checking.

CLM is External Load Module (.LO)

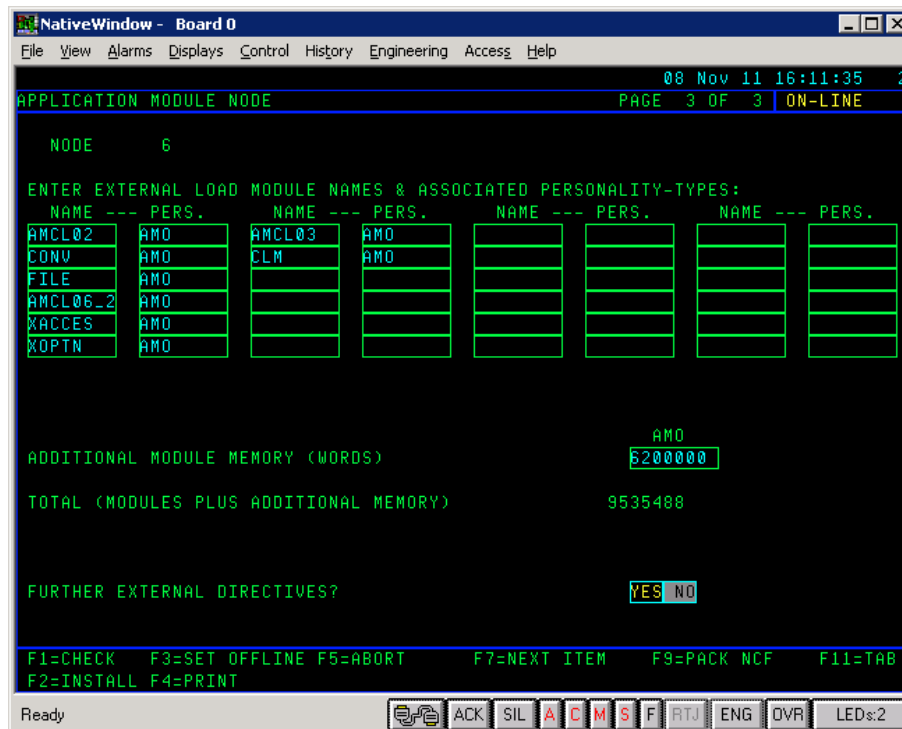
The standard TPN Custom Loader uses the Network Configuration File to load the CLM Platform software.

Note:

- You must include CLM.LO as the last load module while configuring APP/CLM.
- Complete information regarding NCF updates is included in the TPN Implementation/Startup & Reconfiguration - 1, Network Data Entry manual (SW11-305). The following is a summary.

3.5.8.2. Always Backup NCF Before Making Changes

Set up to back up the existing NCF to removable media. Always copy an existing NCF to removable media before installing a new NCF. This allows system recovery without restarting nodes should the new NCF cause problems in loading nodes or the HM autoboot after a power failure.



Setup to Back Up Existing NCF to Removable Media.

Step	Action	Result
1	Select the SUPPORT UTILITIES target from the Engineering Personality Main Menu, with Keylock in the ENGR position.	The Utilities menu displays.
2	Select the MODIFY VOLUME PATHS target.	The Modify Default Volume Path Names Menu displays.
3	Insert the removable medium that will contain your backup NCF into a drive with a known designation.	
4	Find the display entry port for NCF BACKUP PATH.	
5	Enter the drive and volume name of your backup NCF removable media. The format is \$Fn>&ASY> where n = drive number containing the backup NCF – for example, \$F1>&ASY>	The backup pathname is established.
6	Verify that the port for EXT LOAD MODULE has NET>&CUS> in it. Then press ENTER.	The existing NCF is now ready to be backed up to removable media. Actual backup is automatically initiated DURING the NCF update process.
7	To return to the Engineering Personality Main Menu, select MAIN MENU .	Engineering Personality Main Menu displays.

3.5.8.3. Modify NCF

Update the NCF for each new CLM and check for errors. During the processing, the system automatically makes a working copy and a backup copy of the existing NCF; then it updates the working copy. If this process is successful, the system then updates the backup copy also.

This procedure must be repeated for each CLM Node in your system.

Step	Action	Result
1	Select the LCN NODES target, from the Engineering Personality Main Menu, with Keylock in the ENGR position.	This invokes the Network Configurator and displays a list of all the LCN Node numbers presently configured.
Note: Make sure the words ON LINE appear as the indicated mode (second line from the top). If they do not, refer to the <i>Network Data Entry</i> manual in the <i>Startup and Reconfiguration - 1</i> binder to find out why you are not in on-line mode.		
2	Select the node number of the AM to be configured as a CLM.	
3	Select the MODIFY NODE target on display page 1.	Page 2 is displayed.
Step	Action	Result
4	Change the number of Background CL tasks as needed (not required for normal CLM operation). Caution: The CL Background tasks are not required for normal CLM operation. Care must be taken to not exceed the total number of tasks available. The CLM uses 9 of these tasks, Data Access uses 4, and certain AM external load modules may use 1. Prior to LCN R430, this cumulative value must be <= 20. For LCN R43x, this cumulative value must be <= 25. For LCN R500 and later, this cumulative value must be <= 30.	
5	Change the Concurrent Data Access" value to 4. See caution in step 4.	
6	Change the REDUNDANT BUFFER INCREASE (# of 32 KW blocks) to at least 10, if redundant CLMs are implemented. Leave at 0 if non-redundant.	
Note: Some included CL modules execute in the background insertion order. If these CL blocks are used, then the CL Background tasks must be set to an appropriate value and factored in to the total number of tasks calculated.		

Step	Action	Result
7	Page forward to page 3.	Page 3 displays.
8	If the module CONV is not already included then it must be put in the 1st slot/entry location using the personality of AMO. In the next available slot/entry, add the module CLM with the personality of AMO. After these modifications are entered, press ENTER to effect the changes.	The indicated mode at the second line from the top is replaced with the indication that a modification is pending ("MOD AM ##" in white).
Note: Configuration changes must be checked (by pressing the F1 key) and installed (by pressing the F2 key and then pressing ENTER). The Network Configurator does not allow any other changes until you either install the data, OR abort the change by pressing the F5 key. Therefore, complete this procedure by performing the following steps.		
9	Press the Check Key (F1) to check the syntax of the configuration update.	If the syntax check is successful, the AM/CLM Node name (in the mode indicator location on the second line) changes from white to yellow. If the syntax check fails, it changes to red.
Note: If the F1 check fails, correct and repeat the step.		
10	Press the Install Key (F2) and then press the ENTER key to install the configuration update.	The NCF update is installed.
Note: Wait for the INSTALLATION COMPLETE message. The existing NCF file is copied to removable media. The working copy becomes the new NCF.		
11	Press CTL/MENU to complete this step and terminate the NCF updating.	The update is complete.

ATTENTION

Reference Section 9 in the *CLM Platform Implementation Manual* for CLM Redundancy/Automatic Failover NCF information.

3.5.9. Restart the APP with CLM Platform

This procedure must be performed for each **APP/CLM** Node being installed or modified in the system. This will result in the APP being loaded as a **APP/CLM** .

Step	Action	Result
1	Shut down the APP.	
2	Reset the LCNP.	This must be done prior to loading the Native AM personality from the LCN. Caution! Check the Services function to be sure that the TDC Emulators Services is STARTED!
3	Restart the APP.	The CLM Platform software is loaded as part of the standard AM load. Upon completion of the software load to the CLM Node, the Custom Loader activates the CLM Startup task. When the CLM Startup task completes its functions, the CLM Node is in a normal operating state.
4	Check the HM journal entries for errors.	If the CLMI configuration file is incomplete or was not accomplished previously, an error condition is identified in the operator messages entry file.

3.5.10. Adding Other APP/CLM after the First Installation

3.5.10.1. Adding other CLMs

The procedure to add additional **APP/CLM** nodes after the original installation of the **APP/CLM** Platform product can be accomplished by following the installation procedures defined.

Only a portion of the process is required. Start with the procedure Build Points and perform all subsequent procedures up to and including the procedure to Load the APP with CLM Platform Software.

However, when compiling the custom displays, Step 9 is not necessary when adding another CLM. Only the CLMSTAT1 display needs to be modified and compiled.

In summary, to add another **APP/CLM** to your system:

Step	Action
1	Build the CLM_NN and BUILD_NN points.
2	Checkpoint the CLM.
3	Modify and compile the CLM Status display/schematic.
4	Build the required CLMI directive file.
5	Modify the NCF parameters.
6	Restart the CLM Node.

3.6. Install CLM Application

At this point, both parts of the **APP/CLM** Platform are installed. Any further installation process will be application specific.

To install an application, the platform functionality BUILDER is required. Now that installation is complete, BUILDER is available for use. This is the end of **APP/CLM** platform-specific installation.

Proceed to the protocol specific user's manual to install the Communication Protocol and the Network Interface Layer of the CLM and the Windows interface.

4. Interface Errors

System software indications that indicate hardware failures are presented in the Alarm Summary and Message Summary display, as well as in Custom schematics.

A complete list of the error messages and meanings is supplied in the *Process Operations Manual* for the CLM.

5. Appendix – UCLM: CCP AND UCIM Redesigned

CCP (51402426) and UCIM (51402436) boards have been redesigned and are now known as UCLM-51306409. UCLM is a combination of CCP and UCIM.

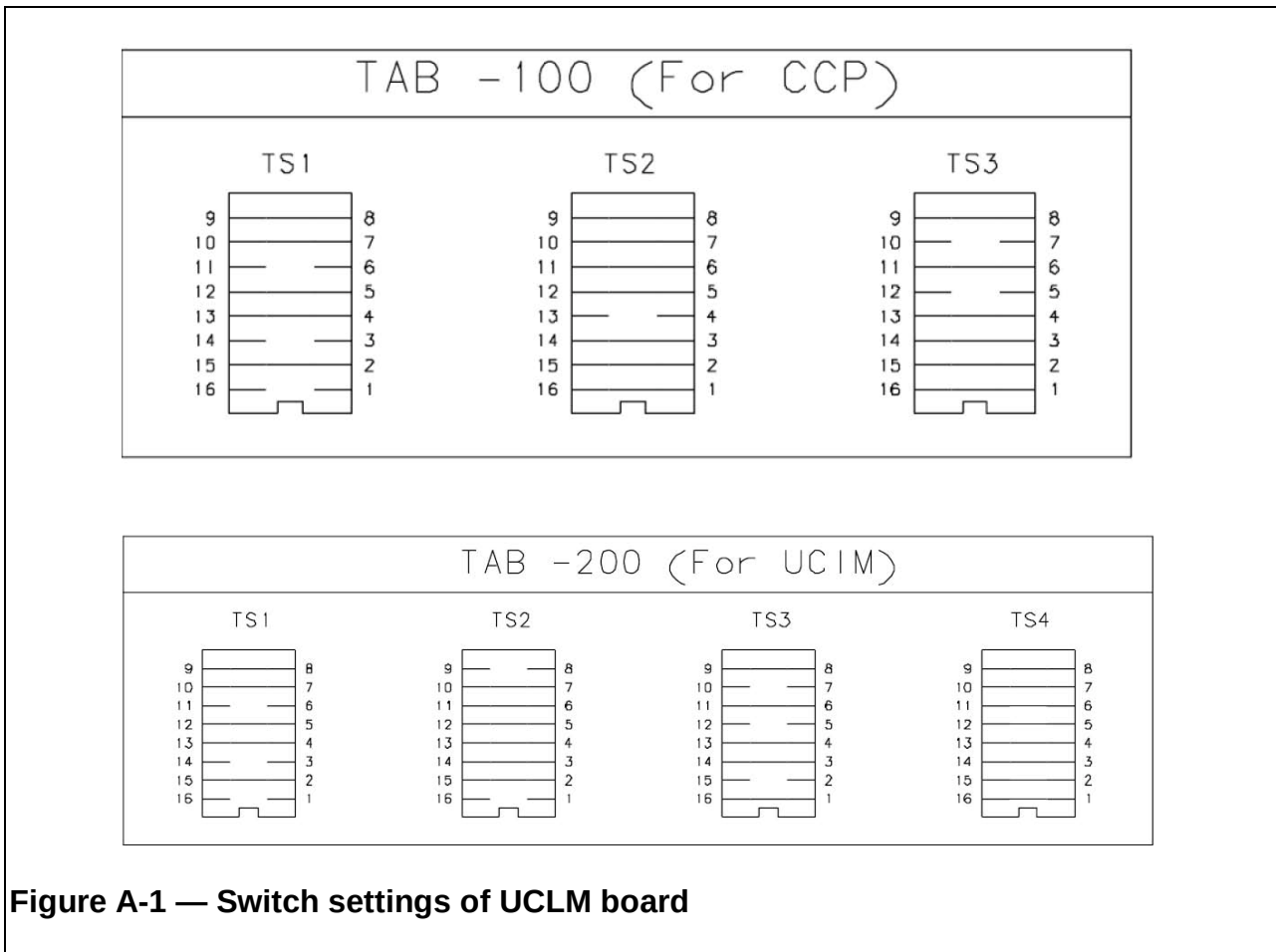
5.1. Ordering Information

To order the boards use the Honeywell part numbers as given below:

Part Number	Board Name
51306409-100	CCP
51306409-200	UCIM

5.2. Switch Settings of UCLM board

Figure A-1 illustrates the redesigned CCP and UCIM settings.



There is no change in the procedure to load AM personality on UCLM. Procedure remains the same as it was in CCP and UCIM.

Honeywell

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA